

Biotechnology and Food Engineering

Process Biotechnology

Spring 2026

Tutorial #2 – growth medium and growth kinetics –

Solutions

Question 1

A pharmaceutical factory tested 2 bacterial strains which can produce a certain substance that has medical activity. The following table describes the data obtained in the test:

	Strain A	Strain B
Initial cell count per liter	100	100
Final cell count per liter	$1 \cdot 10^9$	$5 \cdot 10^8$
Logarithmic growth time	10 hours	9 hours
$Y_{x/s}$ (gr biomass/gr substrate)	0.48	0.42
$Y_{p/x}$ (mg product/gr biomass)	150	120

It can be assumed that in 1 gr of biomass there are $5 \cdot 10^7$ bacterial cells.

- A. What is the growth rate of each strain? *(strain A 1.61 hr^{-1} , strain B 1.71 hr^{-1})*
- B. If we want to produce 1 kg of product in each fermentation cycle, what is the required working volume of the fermenter for each strain?
(strain A 333.33 liter, strain B 833.88 liter)
- C. How much glucose will be consumed per fermentation cycle (based on what was calculated in the previous section) for each strain?
(strain A 13.88 kg, strain B 19.84 kg)

Solution:

A. $\mu = ?$

The growth rate can be extract from the logarithmic growth equation:

$$\ln\left(\frac{X_2}{X_1}\right) = \mu \cdot \Delta t$$

$$\begin{array}{c} \textbf{Strain A} \\ \ln\left(\frac{1 \cdot 10^9 \frac{\text{cells}}{L}}{100 \frac{\text{cells}}{L}}\right) = \mu_A \cdot 10hr \\ \mu_A = 1.61 \frac{1}{hr} \end{array}$$

$$\begin{array}{c} \textbf{Strain B} \\ \ln\left(\frac{5 \cdot 10^8 \frac{\text{cells}}{L}}{100 \frac{\text{cells}}{L}}\right) = \mu_B \cdot 9hr \\ \mu_B = 1.71 \frac{1}{hr} \end{array}$$

B. $V = ?$

Given: $\Delta W_p = 1Kg_p$

Solution rational:

The cells concentration is given in the question.

If we will find the required amount of cells, we can find the working volume of each fermenter.

We will use $Y_{\frac{p}{x}}$, given in the question, to find the amount of biomass required from each strain to produce 1 kg product:

$$Y_{\frac{p}{x}} = \frac{\Delta W_p}{\Delta W_x}$$

$$\begin{array}{c} \textbf{Strain A} \\ 150 \frac{mg_p}{g_{x_A}} = \frac{1Kg_p \cdot 10^6 \frac{mg_p}{Kg_p}}{\Delta W_{x_A}} \\ \Delta W_{x_A} = 6,666.67 g_{x_A} \end{array}$$

$$\begin{array}{c} \textbf{Strain B} \\ 120 \frac{mg_p}{g_{x_B}} = \frac{1Kg_p \cdot 10^6 \frac{mg_p}{Kg_p}}{\Delta W_{x_B}} \\ \Delta W_{x_B} = 8,333.33 g_{x_B} \end{array}$$

The cells concentration at the end of the fermentation is given in the question, and the correlation between the amount of cells to their weight is $1g_X = 5 \cdot 10^7 \text{ cells}$.

While - $\Delta W_X = V \cdot \Delta X$

<u>Strain A</u> $6,666.67g_{X_A} = V_A \cdot \frac{(1 \cdot 10^9 - 100) \frac{\text{cells}}{L}}{5 \cdot 10^7 \frac{\text{cells}}{g_{X_A}}}$ $V_A = 333.3L$		<u>Strain B</u> $8,333.33g_{X_B} = V_B \cdot \frac{(5 \cdot 10^8 - 100) \frac{\text{cells}}{L}}{5 \cdot 10^7 \frac{\text{cells}}{g_{X_B}}}$ $V_B = 833.3L$
--	--	--

C. $\Delta W_S = ?$

The weight of glucose consumed by each strain can be calculated by the growth yield:

$$\Delta Y_{\frac{X}{S}} = \frac{\Delta W_X}{\Delta W_S}$$

While the growth yield is given in the question and the biomass weight was calculated in section B.

<u>Strain A</u> $0.48 \frac{g_{X_A}}{g_S} = \frac{6,666.67g_{X_A}}{\Delta W_S}$ $\Delta W_S = 13,888.9g_S$ $\Delta W_S = 13.88Kg_S$		<u>Strain B</u> $0.42 \frac{g_{X_B}}{g_S} = \frac{8,333.33g_{X_B}}{\Delta W_S}$ $\Delta W_S = 19,841.27g_S$ $\Delta W_S = 19.84Kg_S$
---	--	--

Question 2

A factory is performing a batch fermentation of bacteria for the purpose of biomass production in a 1 m³ fermenter. The bacteria grow on salt medium consists of 2 carbon sources: glucose in an initial concentration of 2% (w/v), and methanol. The fermenter was inoculated with 10 liter starter culture (or simply – starter) in a concentration of 1% bacteria, and the bacteria immediately begin logarithmic growth. First, the bacteria utilize the glucose ($Y_{x/s}=50\%$), and after 10 hours of growth the bacteria turn to use the methanol ($Y_{x/s}=0.639 \frac{g\ X}{g\ S}$). The fermentation ends after 16 hours when the bacterial concentration reaches 39.75 gr/lit, and then the fermentation content is transferred to separation processes.

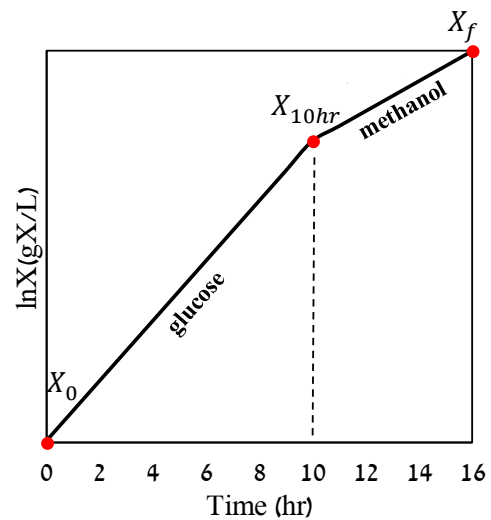
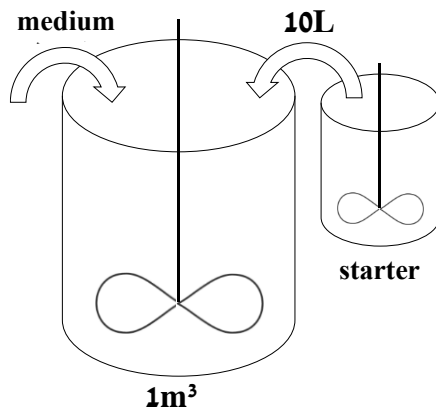
Additional data: methanol density is 769 kg/m³ and the bacteria can exploit 93% of the methanol in the medium. Moreover, the growth rate of the bacteria on methanol is exactly half of the growth rate on glucose.

- A. What is the growth rate of the bacteria on methanol?** *(0.23 hr⁻¹)*
- B. What is the glucose concentration in the medium at the end of the fermentation?** *(0.3 gr/lit)*
- C. What is the required initial methanol volume (in liter) that should be added to the medium while preparing it, before the beginning of the fermentation?** *(65.21 liter)*

Solution:

Given:

<u>Starter</u>	<u>Methanol</u>	<u>Glucose</u>	<u>Fermentation</u>
$V_{\text{starter}} = 10L$	$\rho_M = 769 \frac{KgM}{m^3}$	$S_{0G} = 2\% = 20 \frac{gG}{L}$	$V = 1m^3 = 1000L$
$X_{f\text{starter}} = 1\% = 10 \frac{gX}{L}$	$\Delta S_M = 0.93 S_{0M}$	$Y_{\frac{X}{S_G}} = 50\% = 0.5 \frac{gX}{gG}$	$t_{lag} = 0hr$
	$Y_{\frac{X}{S_M}} = 0.639 \frac{gX}{gM}$	$\Delta t_G = 10hr$	$\Delta t_T = 16hr$
	$X_f = 39.75 \frac{gX}{L}$		
	$\mu_M = 0.5 \cdot \mu_G$		



A. $\mu_M = ?$

The growth rate can be calculated from the logarithmic growth equation:

$$\ln\left(\frac{X_2}{X_1}\right) = \mu \cdot \Delta t$$

A common mistake is to use the growth yield to find X_{10hr} , but pay attention – it is not given in the question that the glucose is completely consumed, and even in section B it is implied that it might not be finished. Therefore, we can't assume that $S_f \approx 0$.

Glucose

It is given in the question that the bacteria turn to methanol after 10 hr, therefore:

$$\Delta t_G = 10hr$$

The biomass origin at the beginning of the growth on glucose is from the starter added to the fermenter:

$$X_0 = \frac{V_{\text{starter}} \cdot X_{f\text{starter}}}{V}$$

$$X_0 = \frac{10L \cdot 10 \frac{gX}{L}}{1000L} = 0.1 \frac{gX}{L}$$

$$\ln \left(\frac{X_{10hr}}{0.1 \frac{gX}{L}} \right) = \mu_G \cdot 10hr$$

Methanol

It is given in the question that the fermentation stops after 16 hr.

$$\begin{aligned} \Delta t_{Ferm} &= \Delta t_G + \Delta t_M \rightarrow \Delta t_M \\ &= \Delta t_{Ferm} - \Delta t_G \end{aligned}$$

Therefore, the growth period on methanol:

$$\Delta t_M = (16 - 10)hr$$

$$\Delta t_M = 6hr$$

The biomass concentration at the beginning of the growth on methanol is the concentration at the end of the growth on glucose, so:

$$\ln \left(\frac{39.75 \frac{gX}{L}}{X_{10hr}} \right) = \mu_M \cdot 6hr$$

It is given in the question that

$$\mu_M = 0.5 \cdot \mu_G, \text{ therefore:}$$

$$\ln \left(\frac{39.75 \frac{gX}{L}}{X_{10hr}} \right) = 0.5 \cdot \mu_G \cdot 6hr$$

We get 2 equations with 2 unknowns:

$$\ln\left(39.75 \frac{gX}{L}\right) - \ln(X_{10hr}) = \mu_G \cdot 3hr$$

$$\ln(X_{10hr}) - \ln\left(0.1 \frac{gX}{L}\right) = \mu_G \cdot 10hr$$

By summing these 2 equations we get:

$$\ln\left(39.75 \frac{gX}{L}\right) - \ln\left(0.1 \frac{gX}{L}\right) = \ln\left(\frac{39.75 \frac{gX}{L}}{0.1 \frac{gX}{L}}\right) = \mu_G \cdot 13hr$$

$$\mu_G = 0.46 \frac{1}{hr}$$

Given in the question that $\mu_M = 0.5 \cdot \mu_G$ so:

$$\mu_M = 0.5 \cdot 0.46 \frac{1}{hr}$$

$$\boxed{\mu_M = 0.23 \frac{1}{hr}}$$

B. $S_{fG} = ?$

The final glucose concentration at the end of the fermentation we can find from the growth yield on glucose:

$$Y_{\frac{X}{S}} = \frac{\Delta X}{\Delta S} = \frac{X_1 - X_0}{S_0 - S_f}$$

The glucose concentration at the beginning of the growth and the growth yield on glucose are given in the question, and X_0 was calculated in section A. We need the biomass concentration at the end of the growth on glucose, X_{10hr} , which can be calculated from the growth equation we developed in section A. by the growth equation on glucose:

$$\ln(X_{10hr}) - \ln(0.1) = 0.46 \frac{1}{hr} \cdot 10hr$$

$$X_{10hr} = 9.95 \frac{gX}{L}$$

By placing numbers in the growth equation we get:

$$Y_{\frac{X}{S_G}} = \frac{X_{10hr} - X_0}{S_{0G} - S_{fG}}$$

$$0.5 \frac{gX}{gG} = \frac{(9.95 - 0.1) \frac{gX}{L}}{(20 - S_{fG}) \frac{gG}{L}}$$

$$\boxed{S_{fG} = 0.3 \frac{gG}{L}}$$

C. $V_M = ?$

To find the initial methanol volume required in the medium before the beginning of the growth, we have to first find the amount of methanol consumed during the fermentation, which can be calculated from the growth yield on methanol:

$$Y_{\frac{X}{S_M}} = \frac{\Delta X}{\Delta S} = \frac{X_f - X_{10hr}}{\Delta S_M}$$

The biomass concentration at the end of the growth on glucose (which is the biomass concentration at the beginning of the growth on methanol) we calculated in section B.

The biomass concentration at the end of the growth and the growth yield on methanol are given in the question. It is also given that 93% of the methanol is consumed, therefore:

$$0.639 \frac{gX}{gM} = \frac{(39.75 - 9.95) \frac{gX}{L}}{(0.93 \cdot S_{0M}) \frac{gM}{L}}$$

$$S_{0M} = 50.14 \frac{gM}{L}$$

The initial amount we will calculate by the volume of the fermenter:

$$\begin{aligned} W_{S_{0M}} &= V \cdot S_{0M} \\ W_{S_{0M}} &= 1000L \cdot 50.14 \frac{gM}{L} \\ W_{S_{0M}} &= 50.14KgM \end{aligned}$$

We will use the methanol density in order to find the required methanol volume:

$$V_M = \frac{W_{S_{0M}}}{\rho_M} = \frac{50.14KgM \cdot 1000 \frac{L_M}{m_M^3}}{769 \frac{KgM}{m_M^3}}$$

$V_M = 65.21 L_M$

Question 3

A factory performs a batch fermentation of yeast for the purpose of a certain extracellular protein production, in a fermenter with working volume of 2.2 m³. The growth medium consists of molasses as the carbon source (67.8% solids, 78% of it are sugar), and a salt solution. After sterilization of the medium, a starting culture is added (to complete to the final working volume), and after a lag phase of 1 hour, the yeast start logarithmic growth of 10 hours, while the doubling time of the yeast at that phase is 1.5 hours. At the end of the growth, the fermenter is discharged and its content proceeds to separation processes to obtain the pure protein. The time needed for discharging the fermenter content, cleaning it and sterilization of the medium for the next batch is 3 hours.

The separation process yield is 75.75% and the weekly production of the factory is 20 kg pure protein per week. The factory is working continuously, 7 days a week, 24 hours a day.

Additional data: the bacterial starter culture, which is inoculated to the fermenter at the beginning of the process, comes from starter fermenter with volume of 200 liter and yeast concentration of 4 gr per liter. The yeast yield on the molasses sugar is 0.52 gr yeast to gr

sugar, and it can be assumed that the sugar is completely consumed. The protein production yield is 40 mg protein to gr yeast.

What is the volume of the salts solution which should be added to the fermenter while preparing the fermentation medium, before its sterilization? (1998.9 liter)

Solution:

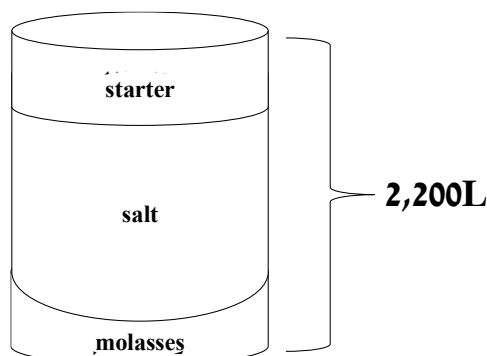
Given:

<u>Yeast</u>	<u>Molasses</u>	<u>Starter</u>	<u>Fermentation</u>
$t_d = 1.5hr$	$0.678 \frac{g_{solids}}{g_{molasses}}$	$V_{starter} = 100L$	$P_0 = 0 \frac{gP}{L}$
$Y_{\frac{X}{Sugar}} = 0.52 \frac{gX}{g_{sugar}}$	$0.78 \frac{g_{sugar}}{g_{solids}}$	$X_{fstarter} = 4 \frac{gX}{L}$	$V = 2.2m^3 = 2200L$
$Y_{\frac{P}{X}} = 40 \frac{mgP}{gX} = 0.04 \frac{gP}{gX}$	$W_{Sf} = 0 g_{sugar}$		$t_{lag} = 1hr$
			$t_{log} = 10hr$
			$t_m = 3hr$
			$W_{pure} = 20 \frac{KgP_{pure}}{week}$
			$\eta = 0.7575 \frac{KgP_{pure}}{KgP}$

A. $V_{salt} = ?$

The fermenter volume includes the molasses volume, salt volume and the starter volume:

$$V_T = V_{molasses} + V_{salt} + V_{starter}$$



The fermenter volume is given in the question, so in order to find the salt volume we need to calculate the volume of each of the other components in the medium.

We will start by finding the **molasses volume** in the fermenter.

Solution rational:

1. By the molasses composition it is possible to calculate its volume. To do that, we need the amount of sugar consumed during the fermentation.
2. We can use the growth yield to calculate the amount of sugar. To do that, we need to find the amount of biomass required to produce the desired protein amount.
3. The required biomass can be calculated from the production yield (product yield on the biomass). To do that, we need the amount of proteins produced in each fermentation.
4. From the weekly production of clean proteins, we can calculate the amount of protein produced in each fermentation.

The amount of product obtained in every fermentation will be calculated from the weekly production. To do that we need the total time of each fermentation:

$$t_{ferm} = t_{lag} + t_{log} + t_m = (1 + 10 + 3)hr = 14hr$$

It is given that the factory works 24 hours a day, 7 days a week, therefore, we can calculate the number of fermentations perform every week:

$$n = \frac{24 \frac{hr}{day} \cdot 7 \frac{day}{week}}{14 \frac{hr}{Ferm}} = 12 \frac{Ferm}{week}$$

The question details that the factory productivity is 20 kg clean protein per week and that the separation processes yield is 75.75%. Therefore, the amount of product **produced** in each fermentation:

$$W_{P_f} = \frac{W_{p_{pure}}}{n} = \frac{20 \frac{KgP_{pure}}{week}}{12 \frac{Ferm}{week} \cdot 0.7575 \frac{KgP_{pure}}{KgP}} = 2.2 \frac{KgP}{Ferm}$$

From the production yield (given in the question) we can find the required biomass weight. Also detailed in the question is that the protein is secreted so its initial concentration in the medium is 0, therefore:

$$Y_{\frac{p}{x}} = \frac{\Delta W_p}{\Delta W_x} = \frac{W_{P_f} - W_{P_0}}{\Delta W_x}$$

$$0.04 \frac{KgP}{KgX} = \frac{(2.2 - 0)KgP}{\Delta W_x}$$

$$\Delta W_x = 55 KgX$$

From the growth yield (given in the question) we can find the amount of molasses sugar that is required in the fermentation. The question also tells us that we can assume that the sugar is completely consumed during the fermentation. Therefore:

$$Y_{\frac{x}{s}} = \frac{\Delta W_x}{\Delta W_s} = \frac{\Delta W_x}{W_{S_0} - W_{S_f}}$$

$$0.52 \frac{KgX}{Kg_{sugar}} = \frac{55KgX}{(W_{S_0} - 0)Kg_{sugar}}$$

$$W_{S_0} = 105.77 Kg_{sugar}$$

In order to find the amount of molasses that should be added to the fermenter (to provide the amount of sugar calculated before) we will use the data given on the molasses composition:

$$W_{S_0 \text{ molasses}} = \frac{105.77Kg_{sugar}}{0.678 \frac{Kg_{solid}}{Kg_{molasses}} \cdot 0.78 \frac{Kg_{sugar}}{Kg_{solid}}} = 200 Kg_{molasses}$$

We know from the question that the molasses consists of 67.8% solids, which means it consists of 32.2% water. As the molasses solids will be fully dissolved in the medium and will not have any volume - the volume given by the molasses comes only from its water content, so the molasses volume in the fermenter is:

$$V_{\text{molasses}} = \frac{W_{S_0 \text{ molasses}} \cdot \frac{Kg_{\text{water}}}{Kg_{\text{molasses}}}}{\rho_{\text{water}}}$$

$$V_{\text{molasses}} = \frac{200Kg_{\text{molasses}} \cdot 0.322 \frac{Kg_{\text{water}}}{Kg_{\text{molasses}}}}{1 \frac{Kg_{\text{water}}}{L_{\text{water}}}}$$

$$V_{\text{molasses}} = \mathbf{64.4 L}$$

Now we only need to find the **starter volume**. The cells concentration in the starter fermenter is given in the question, so in order to find the starter volume that was transferred to the production fermenter we need to find the number of cells in the production fermenter at the beginning of the fermentation. from the logarithmic growth equation:

$$\ln\left(\frac{W_{x_2}}{W_{x_1}}\right) = \mu \cdot \Delta t$$

While the logarithmic growth period is given in the question, we need to calculate the growth rate by using the doubling time:

$$\mu = \frac{\ln(2)}{t_d} = \frac{\ln(2)}{1.5hr} = 0.462 \frac{1}{hr}$$

Therefore:

$$\ln\left(\frac{W_{x_f}}{W_{x_0}}\right) = 0.462 \frac{1}{hr} \cdot 10hr$$

$$\frac{W_{x_f}}{W_{x_0}} = 101.49 \rightarrow W_{x_f} = 101.49 \cdot W_{x_0}$$

For the calculation of the molasses volume we found that $\Delta W_X = 55Kg_X$ meaning that $W_{X_f} - W_{X_0} = 55Kg_X$. Therefore, when we place W_{X_0} instead of W_{X_f} :

$$101.49 \cdot W_{X_0} - W_{X_0} = 55Kg_X$$

$$W_{X_0} = 0.547Kg_X = 547g_X$$

The source for the initial cells weight is in the starter volume added to the fermenter. Given in the question that the cells concentration in the starter is $4 \frac{g_X}{L}$. From here, the starter volume:

$$W_{X_0} = V_{\text{starter}} \cdot X_{f_{\text{starter}}}$$

$$547g_X = V_{\text{starter}} \cdot 4 \frac{g_X}{L}$$

$$V_{\text{starter}} = \mathbf{136.7 L}$$

Therefore, the salt solution volume that should be added to the fermenter before its sterilization is:

$$V_{\text{salt}} = V_T - (V_{\text{molasses}} + V_{\text{starter}})$$

$$V_{\text{salt}} = 2200L - (64.4 + 136.7)L$$

$$\boxed{V_{\text{salt}} = 1998.9 L}$$